

The MapReduce framework consists of a single master JobTracker and one slave TaskTracker per cluster-node. The JobTracker is responsible for tracking resource consumption/availability, scheduling the jobs component tasks on the slaves, resource management, and monitoring them and re-executing the failed tasks. The slave TaskTrackers execute the tasks assigned by the master and provide task-status information to the master periodically.

Economics of Hadoop

Big Data comes at Big Costs. Hadoop is not the only Big Data platform in the industry, but the software has created a buzz around in a short span of time. For Hadoop, the biggest motivator in the market is simple: Before Hadoop, data storage was expensive. Hadoop lets you store any amount of data, structured or unstructured, simply by adding more servers to a Hadoop cluster. These servers are x86 based machines which come at relatively low cost and add more processing power and more storage to the overall cluster. This makes data storage with Hadoop far less costly than its competitors.

“The cost of a Hadoop data management system, including hardware, software, and other expenses, comes to about \$1,000 a terabyte--about one-fifth to one-twentieth the cost of other data management technologies,” says Charles Zedlewski, VP of product at Cloudera. If we were to store the same terabyte of data at a network storage, it wouldn't be unreasonable to think of a price of \$5,000 per terabyte or even more. Moreover, if we were to also include the hardware and other physical resources involved with the network storage, the cost would sky rocket to not less than \$10,000-\$15,000 per terabyte. On the other hand, if we thought of still sticking to the legacy data management technologies like RDBMS, the total cost might be more like \$30,000 to \$40,000 per terabyte.

Forrester Research coined the term “Hadooponomics”. The main reason for Hadoop being huge in business for 2015 is its economical viability. According to Forrester “Hadooponomics” will definitely make adoption mandatory because Hadoop is simply a cheaper way to form large-scale storage repository of data and conduct analytics queries. Hadoop can scale storage and processing and leverage the cloud. The costs for large Hadoop distributions amount to 2,000 -3,000 dollars per node per year. In contrast, a SAP HANA node costs approximately 750,000 dollars per year. A renowned UK company compared the costs of conventional data storage with the estimated costs for a Hadoop replacement. One TB of data in an Oracle database generates costs in the amount of 48,000 Euro per